

VAMSEE ACHANTA, P.E.

Naval Architect | Subsea & Marine Systems Engineering Leader | Digital Transformation Executive

LinkedIn: [linkedin.com/in/vamseeachanta](https://www.linkedin.com/in/vamseeachanta) | **GitHub:** github.com/vamseeachanta | **Portfolio:** [career site](#)

Professional Profile

Marine/offshore engineering leader with 23+ years of experience in naval architecture, subsea systems, risers, pipelines, umbilicals, moorings, installation engineering, integrity management, and engineering automation. Professional Engineer in Texas with a track record leading high-consequence offshore work from concept/FEED through detailed analysis, installation support, operations, and emergency response.

Combines hands-on technical authority in OrcaFlex, AQWA, OrcaWave, ANSYS, Abaqus, and COMSOL with Python/SQL automation and data-science experience. Known for converting complex engineering workflows into repeatable digital tools, digital twins, and production-grade analytics that reduce analysis cycle time, improve quality, and support faster project decisions.

Positions

Period	Role	Organization	Focus
Dec 2023 – Present	Naval Architect	Alan McClure & Associates, Houston, TX	Coupled marine global analysis, diffraction, moorings; WoodFibre LNG FST terminal design
Jun 2016 – Present	VP of Engineering / Co-Founder	Frontier Deepwater Appraisal Solutions (FDAS), Houston, TX	Deepwater production concepts, riser/mooring systems, GoM field economics
Jun 2012 – Present	Engineering Lead Consultant	AceEngineer (independent practice), Houston, TX & Remote	Subsea/riser/installation analysis, FEA, fitness-for-service, digital twins, open engineering & AI tooling
Sep 2017 – Dec 2020	Engineering SME, Data Science Team	Occidental Petroleum, Houston, TX & Remote	Physics-based production/drilling analytics across ~20,000 wells
Aug 2003 – Jun 2015	Engineering Lead	2H Offshore Inc, Houston, TX	Riser design & integrity, 100+ projects, BP Macondo containment riser response, API standards



Roles from 2012 onward run concurrently; project-level detail for each position follows on the next pages.

Career Highlights

- **Emergency response leadership:** Engineering Manager for BP Macondo containment riser response; delivered a complete containment riser design in 8 weeks by repurposing existing assets.
- **Major LNG terminal analysis:** WoodFibre LNG Terminal design (British Columbia, \$1.8B) — dual Floating Storage Terminals, shore mooring, LNG carrier berthing/fender interaction, extreme-weather assessment.
- **Subsea and riser authority:** 100+ offshore riser and subsea projects across Gulf of Mexico, West Africa, Asia-Pacific, Guyana, and Venezuela for BP, Shell, Chevron, ExxonMobil, ENI, Murphy, Reliance, Talos, and others.
- **Engineering-workflows platform (Deckhand):** API-first platform with ~30 deterministic, standards-based screening workflows (fatigue, seakeeping, lifting, weather windows) and provenance-gated licensed OrcaFlex/AQWA runs — [live demo gallery](#).
- **Open, demonstrable engineering:** Live public tools — [worldenergydata](#) (GoM field economics on BSEE data) and [digitalmodel](#) (open offshore library) — deterministic, unit-tested, provenance-traced; OTC/OMAE/IADC/World Oil author.

Education & Credentials

Master of Science, Mechanical Engineering — Texas A&M University, College Station, TX

Bachelor of Technology, Mechanical Engineering — Indian Institute of Technology (IIT), Madras, India

Professional Engineer (P.E.) — Texas

API Standards Committee Experience: API-RP-16Q, API-RP-17G, API-RP-17G2

Core Competencies

Marine & Subsea Engineering: Naval architecture, floating systems, FST/FPSO/FPU concepts, floating offshore wind platform sizing, LNG marine operations, risers (SCR, TTR, hybrid, lazy-wave, simple catenary), pipelines, umbilicals, jumpers, manifolds, moorings, CALM/SALM buoys, installation engineering, integrity management, decommissioning.

Analysis & Simulation: Coupled dynamic analysis, diffraction/radiation analysis, mooring analysis, time-domain simulations, global riser analysis, pipelay and installation windows, detailed FEA, nonlinear elastic-plastic analysis, fracture mechanics, fatigue, corrosion simulation, hydrodynamics, operability assessment (DNV-RP-H103 / DNVGL-ST-N001 weather windows and operation envelopes).

CFD: OpenFOAM (RANS $k-\omega$ SST, laminar, free-surface VOF/olaFlow waves) on a license-free FreeCAD/gmsh → OpenFOAM → PyVista pipeline, validated against canonical benchmarks — Blasius flat plate, cylinder $Re=100$ vortex shedding, NACA0012 airfoil, dam-break free surface — with published live validation reports.

Software & Automation: OrcaFlex / OrcFxAPI, AQWA, OrcaWave, ANSYS, Abaqus, COMSOL, Flexcom, Python, SQL, Git, Azure DevOps, pandas, NumPy, Matplotlib, Plotly/D3JS, API-driven model generation, batch execution, automated post-processing, engineering dashboards; AI/LLM-augmented engineering workflows and agentic automation, deterministic zero-server static web delivery (GitHub Pages).

Standards & Governance: API-RP-16Q, API-RP-17G, API-RP-17G2, API 579-1 / ASME FFS-1, BS 7910, DNV offshore recommended practices, ASME VIII, engineering assurance, contractor deliverable review, design basis development, technical risk assessment.

Leadership: Engineering team leadership, mentoring, client-facing technical advisory, project management, contractor coordination, proposal support, business development, emergency response, cross-functional engineering/data-science integration.

Professional Experience & Project Detail

Naval Architect, Alan McClure & Associates

Dec 2023 – Present | Houston, TX

- **WoodFibre LNG Terminal Design — British Columbia (\$1.8B project):** Performed detailed marine global analysis for two Floating Storage Terminals berthed side-by-side in approximately 35 m water depth, including coupled FST/shore-mooring analysis for extreme weather, FST/LNGC/fender/mooring interaction checks for operating conditions, mooring support for LNG carriers up to 180,000 m³, and AQWA/OrcaWave/OrcaFlex workflows for diffraction, hydrodynamics, mooring, and coupled global analysis.
- **Diffraction and mooring failure investigations:** Performed AQWA/OrcaWave diffraction analysis and OrcaFlex global analysis for moored barges to evaluate mooring line failure mechanisms in extreme weather.
- **Tug floundering / passing-ship analysis:** Investigated tug boat floundering due to passing ship effects using time-domain OrcaFlex analysis to assess tug motions and operational risk.
- **Digital transformation and analysis automation:** Developed Python/API-based digital twins for hull diffraction workflows in AQWA/OrcaWave and mooring analysis workflows in OrcaFlex using OrcFxAPI, automated model creation, execution, result extraction, and engineering plot/report generation.

VP of Engineering / Co-Founder, Frontier Deepwater Appraisal Solutions (FDAS)

Jun 2016 – Present | Houston, TX

- Lead concurrent venture-backed engineering and technology development for deepwater field development concepts, riser/mooring systems, field economics, technical marketing, and operator engagement.
- **6000 ft water-depth semisubmersible production vessel concept:** Led feasibility and concept evaluation for converting idle 6th-generation drillships into production assets; delivered riser support, mooring, and hurricane-condition analysis to support cost-effective development decisions.
- **GoM field economics analytics platform:** Built SQL/Python analytics platform using BSEE data for 200+ Gulf of Mexico fields, reducing field evaluation time from weeks to days and supporting stranded-asset opportunity screening.
- **SEWOL salvage ship critical analysis:** Delivered emergency hull FEA analysis within 72 hours for Korean government salvage support; automated ANSYS workflows using Python to accelerate model execution and reporting.
- Developed technical visualizations and digital engineering collateral for operator meetings, proposal support, and strategic business development.

Engineering Lead Consultant, AceEngineer (Independent Consulting Practice)

Jun 2012 – Present | Houston, TX & Remote

- Provide concurrent specialist offshore/subsea engineering consulting for major operators and engineering contractors, covering installation analysis, risers, pipelines, umbilicals, structural FEA, corrosion, and fitness-for-service.
- **Circular BOP design and analysis:** Performed detailed nonlinear and elastic-plastic ANSYS FEA of seals, connectors, annulus seals, drill pipe seals, and metal/elastomer component behavior.
- **Subsea structure installation analysis:** Performed installation analyses for manifolds, PLETs, flexible products, umbilicals, rigid jumpers, and subsea structures.
- **42-inch thin-walled subsea pipeline installation — Venezuela:** Executed detailed FEA to determine bending moment capacity for $D/t = 67$ pipe outside standard DNV code regime; performed OrcaFlex pipelay analysis and automated results/plots via Python/API workflows.
- **ExxonMobil Yellowtail installation analysis — Guyana:** Analyzed dynamic and static umbilical installation in approximately 6,000 ft water depth; evaluated 100+ load cases and installation windows to reduce execution risk.
- **Talos Venice Limerock static umbilical installation:** Assessed static umbilicals from subsea manifold to subsea tree in approximately 1,050 m water depth, including static installation tables and dynamic wave/current limits.
- **Chevron Ballymore installation analysis:** Performed dynamic and resonance analysis for rigid jumpers and manifolds to support installation planning.
- **Riser digital twin platform:** Implemented database-driven automation for drilling risers, top-tensioned risers, lazy-wave catenary risers, simple catenary risers, global riser analysis, API 579 fitness-for-service, and BS 7910 fatigue/fracture mechanics workflows.
- **Trendsetter intervention system:** Supported intervention system design for water depths from 1,500 ft to 10,000 ft and pressures of 15,000–17,500 psi, including global riser analysis, connector/flange specifications, vessel stroke checks, and upper/lower riser configurations.
- **Fitness-for-service automation:** Built Python workflows using pandas, NumPy, Matplotlib, and D3JS for wall-thickness processing, defect identification, API 579-1/ASME FFS-1 assessments, and BS 7910 fracture mechanics.
- **Mooring corrosion simulation:** Modeled damaged subsea mooring line corrosion in COMSOL using fluid mechanics, mass transfer, diffusion, and electrochemical models; automated model processing with COMSOL Java API, MATLAB, and batch workflows.
- **Deckhand engineering-workflows API:** Designed and built an API-first platform (request in → deterministic analysis → report URL out) with a catalog of ~30 standards-based screening workflows — mooring/riser fatigue, spectral fatigue from RAOs, vessel seakeeping, mudmat bearing, lifting lugs, weather windows, liquefaction — versioned algorithm routing, provenance-gated licensed OrcaFlex/AQWA execution on an automated Windows lane, and chat-bot delivery; public [demo gallery](#).
- **Open engineering & AI tooling:** Built and shipped public, deterministic engineering sites and libraries — [worldenergydata](#) (live GoM field economics & well analytics on BSEE data: Lower-Tertiary NPVs, 56-well benchmarking, 3D well paths), [digitalmodel](#) (open offshore engineering library: 221 S-N fatigue curves across 17 standards, OrcaWave/AQWA diffraction benchmarks, vessel/rig database of 2,268 rigs + 165 construction vessels with installation-suitability API, floating-wind platform sizing, rudder/manoeuvring and rod-pump dynacard explorers, validated OpenFOAM CFD suite), and a Raw-to-Knowledge playbook — every result unit-tested and provenance-traced, delivered as zero-server static HTML, with AI/LLM as narrative support over deterministic cores.
- **AI-augmented engineering operations:** Operate a 25+ repository engineering ecosystem with agentic LLM workflows under CI-enforced invariants, including an LLM-assisted solver model-generation pipeline (project data → specification YAML + assumption ledger → licensed OrcaWave/AQWA/OrcaFlex input) where every published number is provenance-gated — LLM proposes, deterministic code disposes.
- **Marine safety / HSE analytics:** Built BSEE incident-database analytics (1,987 GoM incidents across failure modes) powering incident-grounded failure-mode reports and seasonal-preparedness studies for offshore operations.

Engineering SME for Data Science Team, Occidental Petroleum

Sep 2017 – Dec 2020 | Houston, TX & Remote

- Served as domain engineering lead bridging production engineering, drilling, data science, and software teams for large-scale analytics deployment.
- **Production optimization platform:** Deployed 50+ physics-based algorithms monitoring approximately 20,000 wells in real time with logging, exception handling, and production-support workflows.
- **Real-time drilling analytics:** Implemented bit-position estimation, torque/drag monitoring, rig-state logic, and edge-computing workflows supporting 20+ concurrent drilling campaigns.

- **Production surveillance:** Automated dynacard analysis, ESP surveillance, well-test validation, and time-shed analytics to improve exception-based operations and reduce manual review effort.
- Designed SQL data models, stored procedures, Python class libraries, analytics pipelines, dashboards, and validation routines for production-grade engineering analytics.

Engineering Lead, 2H Offshore Inc

Aug 2003 – Jun 2015 | Houston, TX

- Progressed through analyst, lead, manager, and technical authority roles for riser design, subsea systems, installation analysis, integrity management, and API standards work.
 - **BP Macondo emergency response:** Engineering Manager for containment riser response; led 24/7 multidisciplinary engineering effort across multiple locations and delivered complete design in 8 weeks by repurposing existing assets.
 - **Chevron Bangka SCR/flexible riser FEED:** Led FEED design for production risers in shallow-water development, optimizing riser configuration and installation strategy while delivering ahead of schedule.
 - **Major riser and integrity portfolio:** Led or supported BP Thunder Horse, Atlantis, Horn Mountain; Chevron Jack/St. Malo; ENI Devils Tower; Reliance KG-D6; Shell/BP HPHT systems; and multiple Gulf of Mexico, West Africa, and Asia-Pacific developments.
 - **API standards contribution:** Committee member/contributor for API-RP-16Q and related riser/structural standards.
 - Managed client communication, technical reports, design reviews, team coordination, mentorship, and project delivery for 100+ assignments.
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Selected Project Portfolio

- **WoodFibre LNG Terminal:** Dual FST global analysis, LNG carrier berthing, fender/mooring interaction, operational and extreme conditions.
- **BP Macondo containment riser:** Emergency engineering response, rapid design, existing asset repurposing, high-consequence delivery.
- **ExxonMobil Yellowtail:** Dynamic/static umbilical installation, 6,000 ft water depth, installation windows and load-case automation.
- **Chevron Ballymore / Talos Venice Limerock:** Rigid jumper, manifold, and static umbilical installation analysis.
- **42-inch Venezuela pipeline:** Thin-walled pipeline bending capacity and pipelay analysis beyond standard DNV D/t regime.
- **SEWOL salvage:** Rapid-response hull FEA and automated ANSYS reporting for salvage operations.
- **Drilling/riser digital twins:** Automated OrcaFlex model generation, batch execution, results extraction, and API 579 / BS 7910 integrity workflows.
- **Deckhand (live demos):** API-first deterministic engineering-workflows platform — [public demo gallery](#) with grounded, provenance-traced analysis cards.
- **worldenergydata (open):** Live, deterministic Gulf of Mexico field-economics & well-analytics site on public BSEE data — Lower-Tertiary NPVs, 56-well benchmarking, interactive 3D well paths; unit-tested numbers with full provenance.
- **CFD validation suite (open):** OpenFOAM benchmarks validated against canonical solutions — Blasius flat plate, cylinder $Re=100$ vortex shedding, NACA0012, dam-break VOF — with live published reports in digitalmodel.

Selected Publications & Thought Leadership

- **World Oil (2024):** “Wellbay Innovation Supports Lower-Cost HP DW Discoveries”
- **OTC (2017):** “Heavy Lift Dynamics — Sewol Ferry Offshore Salvage”
- **IADC (2015):** Contributing Author, *Floating Drilling Equipment and Operations*, 12th Edition
- **OMAE (2009):** “Benchmarking SHEAR7v4.5: Full-Scale Drilling Riser VIV Data”
- **World Oil Industry Reference (April 2026):** “SF Offshore Technology—Shilling (Frontier Deepwater Appraisal Solutions LLC)”

Professional Affiliations

Society of Petroleum Engineers (SPE) | American Society of Mechanical Engineers (ASME) | Marine Technology Society (MTS) | API Standards Committees

Languages

English — Excellent | Telugu — Excellent | Hindi — Good

References

Available upon request.